

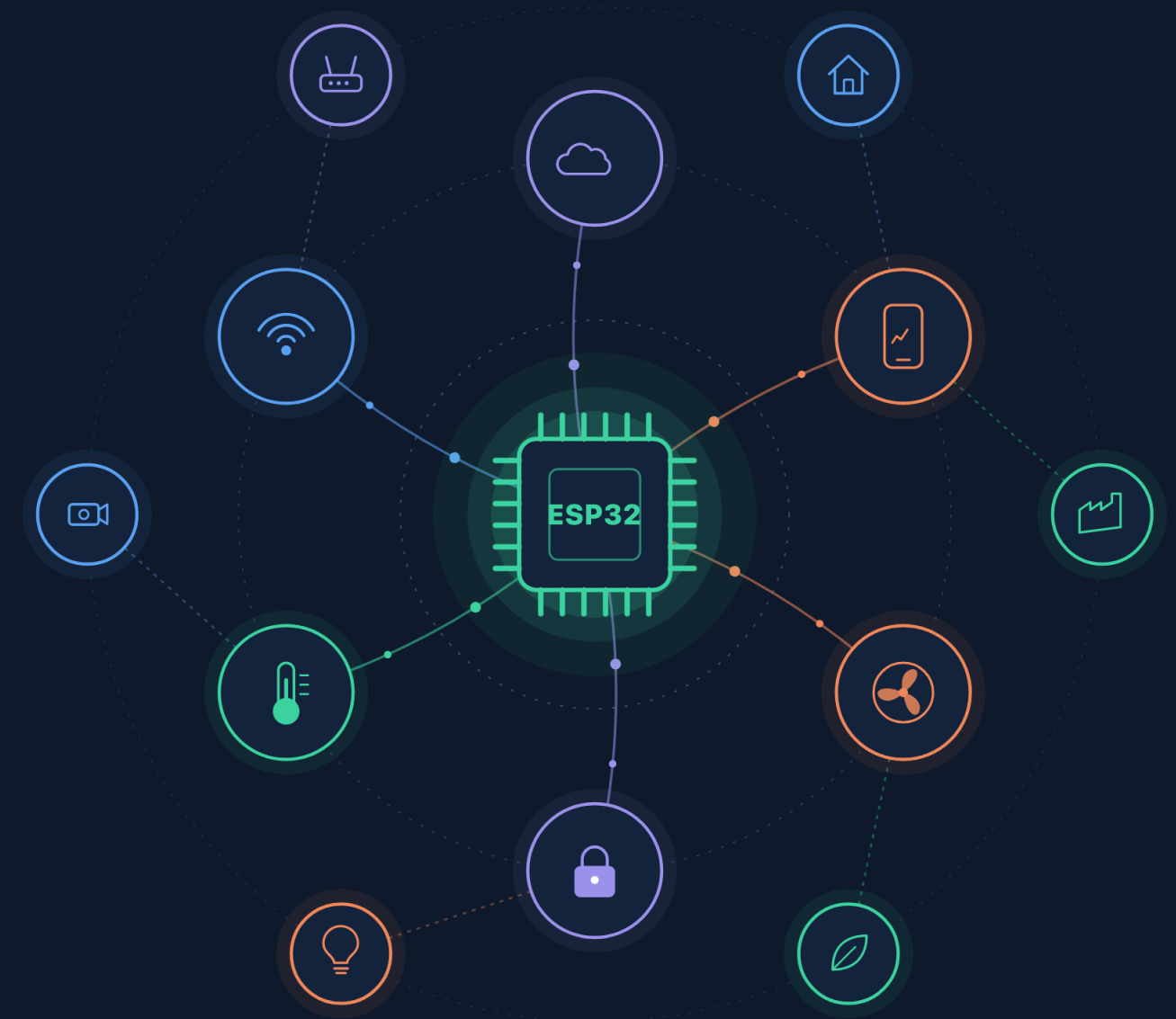
MODULE 2 • 45 MINUTES

How an IoT System Is Built

The five layers inside every IoT system, the small computers that run devices, and the journey of one reading.

Mehdi Lotfinejad

Introduction to the Internet of Things



THE ROAD AHEAD

Eight modules, one smart device

PART 1 — DEVICES AND DATA

01

What is IoT?

Lecture

02

How it's built

You are here

03

Sensors and actuators

Labs 1–2

04

Getting online

Lab 3

PART 2 — CLOUD, AUTOMATION AND SECURITY

05

Cloud and dashboards

Lab 4

06

Automation

Lab 5

07

Security

Group activity

08

Real world + capstone

Lab 6

BY THE END OF THIS MODULE

You will be able to...

-
- 01 **Name** the five layers of an IoT system and what each one does
 - 02 **Explain** the difference between a sensor and an actuator
 - 03 **Compare** a microcontroller with a microprocessor
 - 04 **Trace** one reading from a sensor to a person's screen
 - 05 **Identify** the main groups who make an IoT product happen
-

WHERE WE ARE

Module 1 showed what an IoT system does.
Module 2 shows how it is put together.

Think of lifting the bonnet of a car to see the engine, the wheels and the steering.

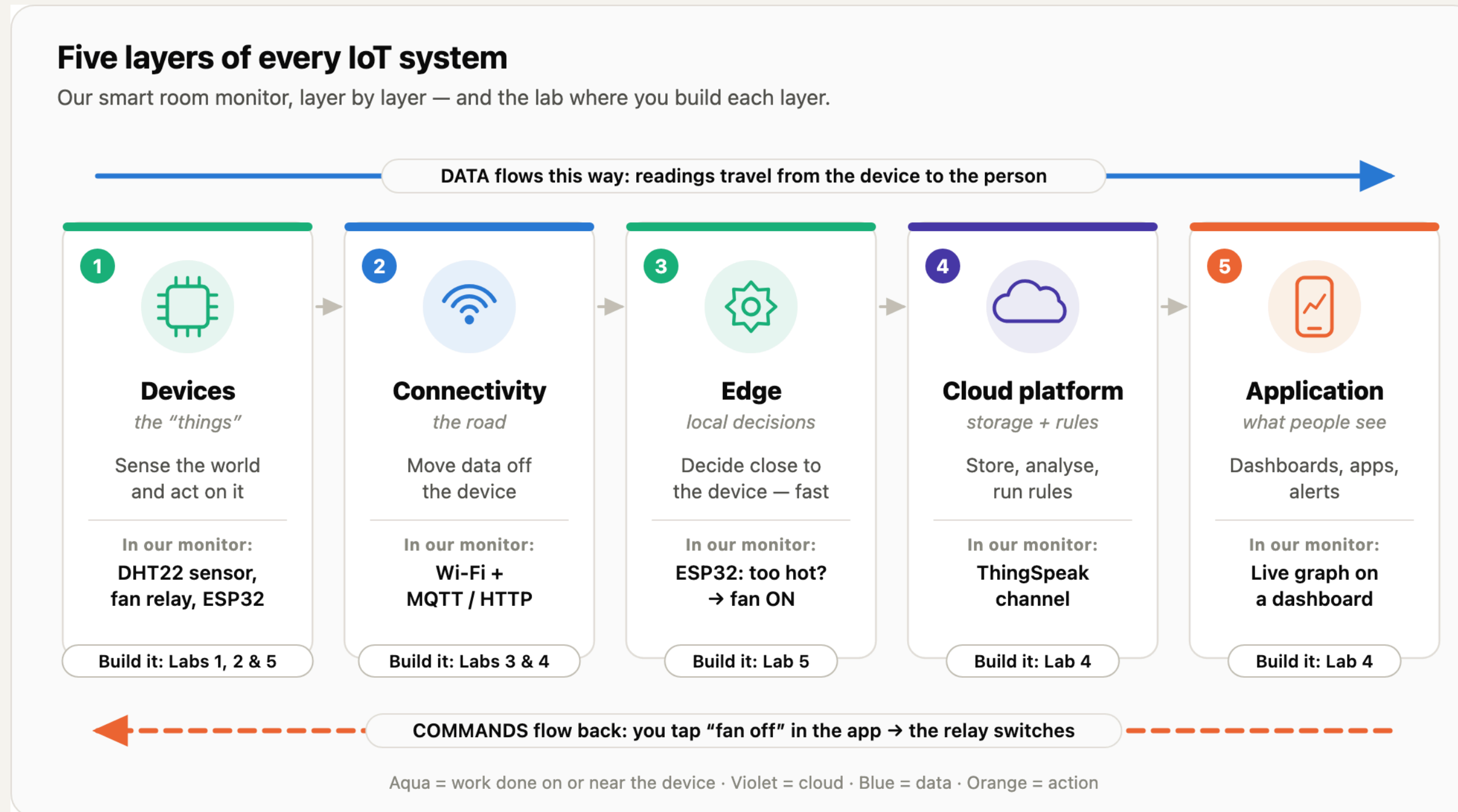
Part A

The five layers

One smart room monitor, broken down layer by layer.

THE BIG PICTURE

Five layers of every IoT system



Three ingredients in every device



Sensors

Measure something in the real world, like temperature.



Actuators

Do something in the real world, like switching on a fan.



A small computer

Usually a microcontroller that runs the program.

In the labs, your device is a simulated **ESP32** board.

The road the data travels on



The radio

How the signal travels, like Wi-Fi or Bluetooth.



The language

What the device speaks, like MQTT or HTTP.



A gateway

A middleman box that serves many nearby devices.

Your home Wi-Fi router is a kind of gateway. So is a smart-home hub.

LAYER 3 · EDGE COMPUTING

Edge computing means doing some of the thinking close to where the data is collected.

On the device itself, or on a gateway nearby — not far away in the cloud. It is faster, and it keeps working when the connection doesn't.

What a cloud platform does



Stores

Keeps old readings,
so you see history.



Processes

Works things out,
like a daily average.



Runs rules

“Above 28 °C? Send
me a message.”



Manages devices

Knows which
devices exist and
may connect.

In Lab 4 you use a free cloud platform, **ThingSpeak**. Module 5 covers this layer.

LAYER 5 · APPLICATION

A farmer doesn't care about data centres. They care that their phone says “Field 3 needs water”.

Dashboards, phone apps, email alerts, reports — or another computer system, like billing.

PAUSE AND CHECK

Our room monitor switches the fan on by itself when it gets hot, without asking the cloud. Which layer is making that decision?

Devices · Connectivity · Edge · Cloud platform · Application

Part B

Inside the device

Senses, muscles and a brain.

Sensors and actuators

	Sensor (input)	Actuator (output)
Job	Measures the world	Changes the world
Like your...	Senses	Muscles
Examples	Thermometer; light, motion and gas sensors	LED, buzzer, motor, relay (an electric switch)
In our monitor	Temperature and humidity sensor	Relay that switches the fan

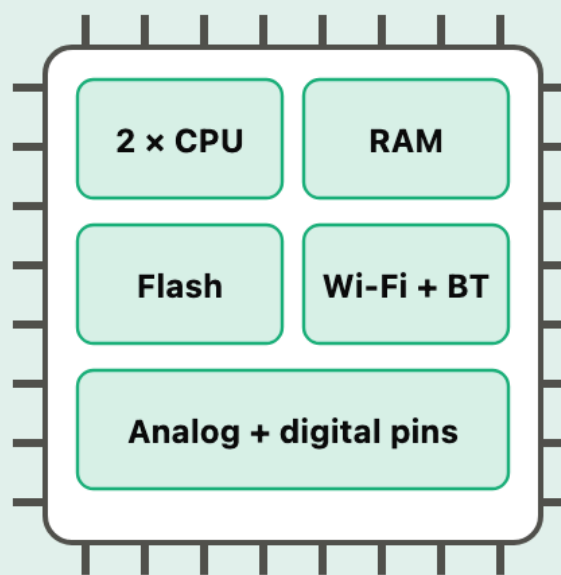
Ask: **which way is the information going?** In from the world = sensor. Out into the world = actuator.

One chip, one job vs a small computer

One chip, one job vs a small computer

A microcontroller packs everything onto one chip. A microprocessor needs a board of helpers and an OS.

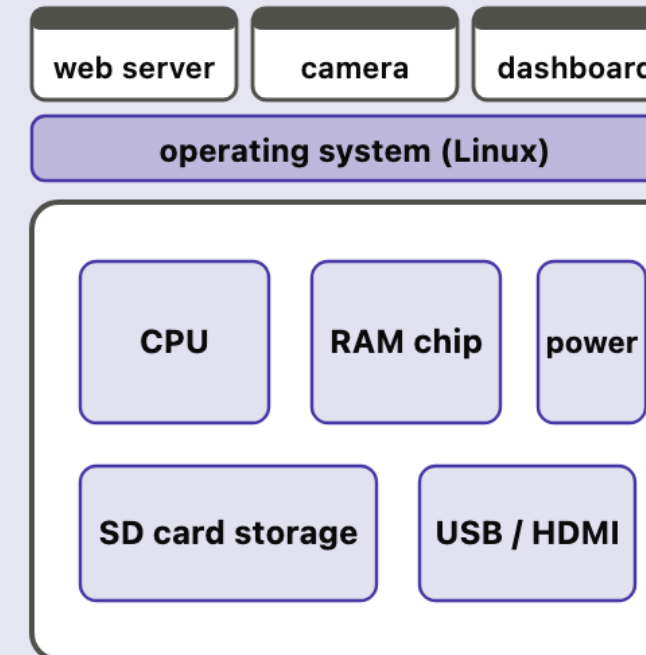
MICROCONTROLLER (e.g. ESP32)



everything on ONE chip

- ✓ **Runs ONE program**,
from the moment power is on
- ✓ **Starts instantly**
no booting
- ✓ **Deep sleep $\approx 10 \mu\text{A}$**
months–years on batteries
- ✓ **Low cost**
cheap enough to put in anything

MICROPROCESSOR BOARD (e.g. Raspberry Pi)



several chips on a board

- i **Runs an OS +**
many programs at once
- i **Boots in seconds**
like a small PC
- i **Draws watts**
usually mains-powered
- i **Great for video, AI**
and heavy software

SIDE BY SIDE

Microcontroller or microprocessor?

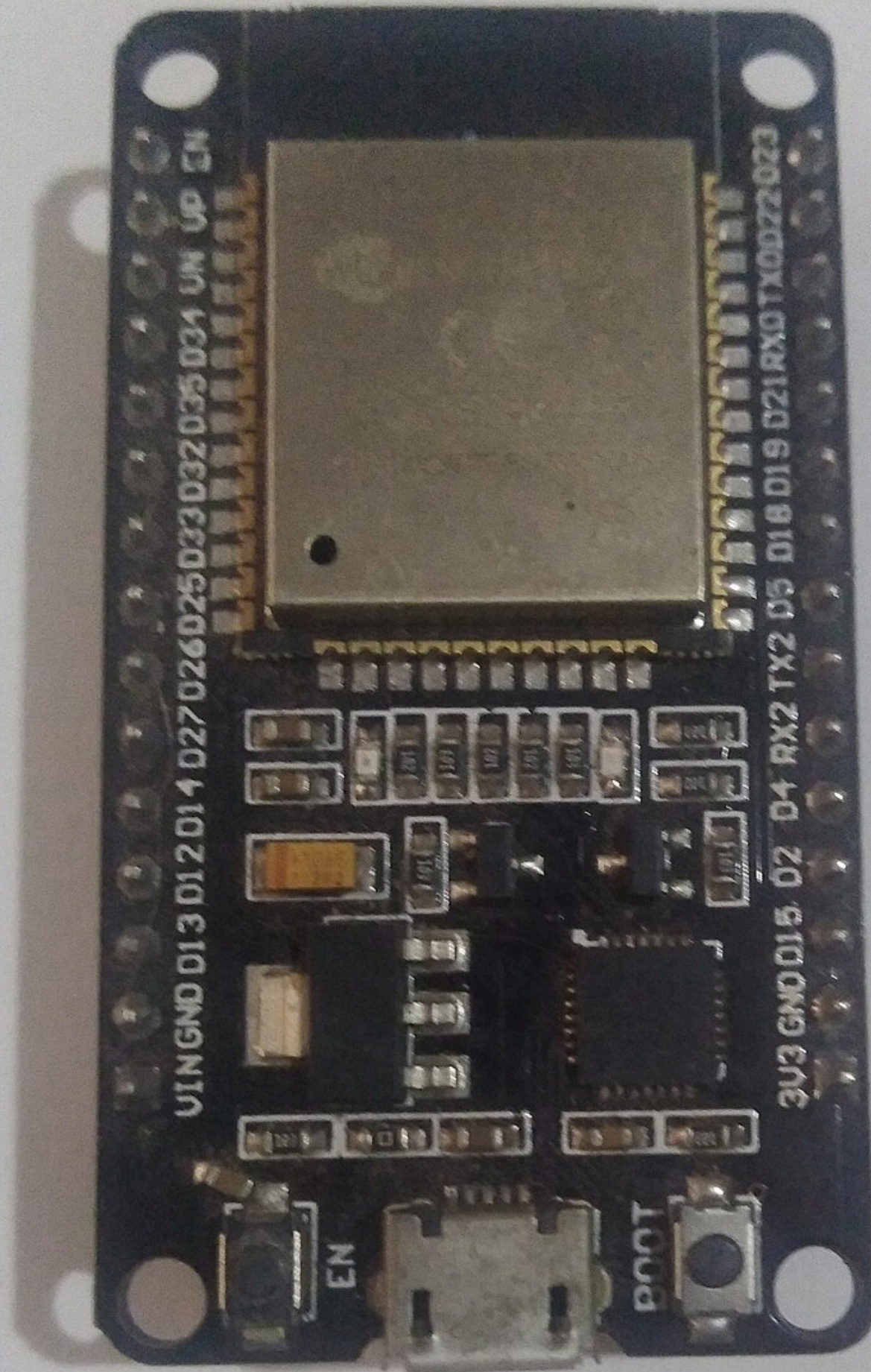
	Microcontroller (MCU)	Microprocessor (MPU)
Like a...	Kitchen timer: one job, done reliably	Laptop: many jobs, more looking after
Runs	One program, straight away	An operating system and many programs
Power use	Very low: months or years on batteries	Higher: usually mains power
Start-up	Almost instant	Seconds to a minute (it boots)
Cost	A few US dollars	Tens of US dollars and up
Good at	Reading sensors, switching things	Video, heavy data, a local dashboard
Examples	ESP32, Arduino boards, STM32	Raspberry Pi, the chip in your phone

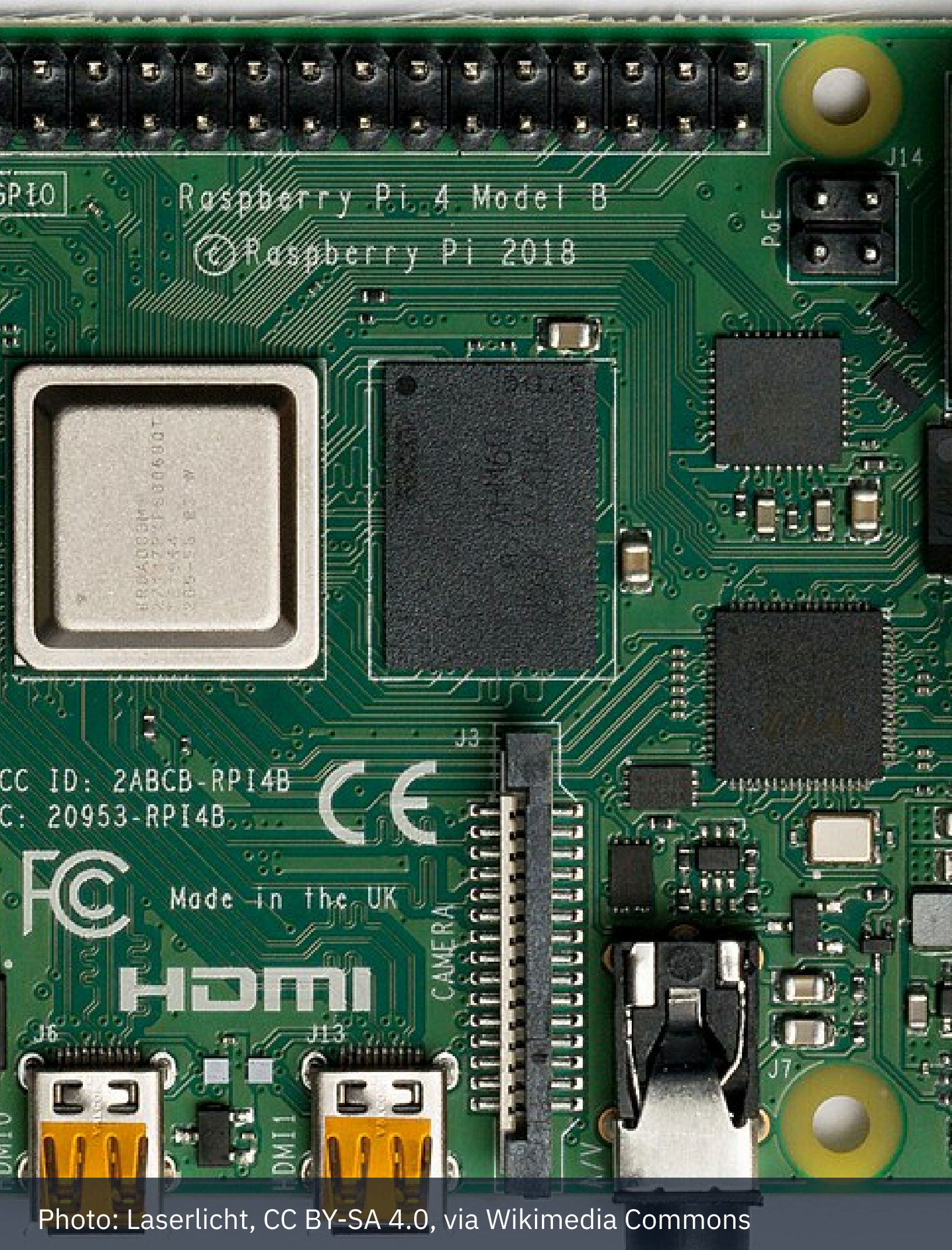
Reads sensors and flips switches? **Microcontroller.** Video or lots of software? **Microprocessor.**

THE DEVICE YOU'LL USE

Meet the ESP32

- **Wi-Fi and Bluetooth** built into the chip
- **Cheap:** a board costs very little
- **Capable:** 2 cores, 240 MHz, 520 KB memory
- **Deep sleep:** the chip uses about 10 microamps
- **Lots of pins** for sensors and actuators





THE OTHER FAMILY

Meet the Raspberry Pi

- Built around a **microprocessor**
- Runs a full **operating system**
- Connects a camera or a screen easily
- Can be the **gateway** for many sensors

Part C

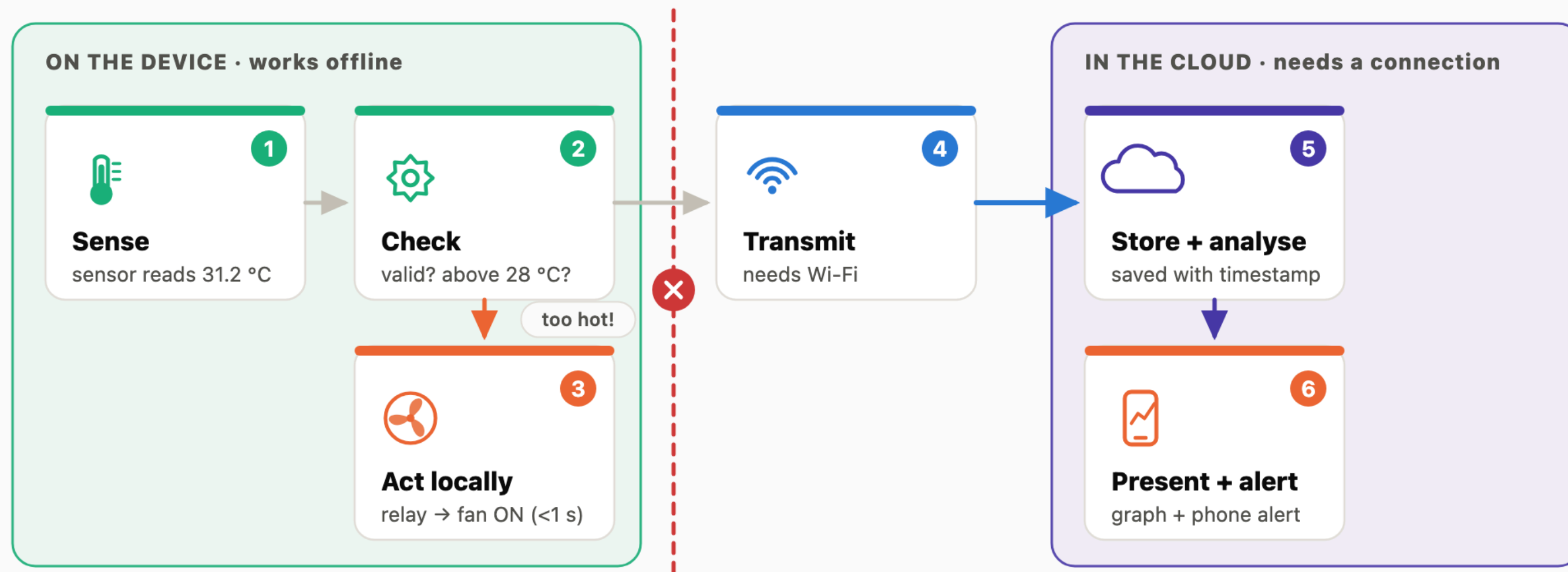
Following one reading

From a sensor, through the layers, to a person — and who makes it all happen.

One reading's journey

Following one reading through the pipeline

31.2 °C is measured. Steps 1–3 happen on the device before any data leaves it.



If Wi-Fi drops, steps 1–3 (left of the red line) still run — the fan still switches on

PAUSE AND CHECK

In the pipeline, which steps would still work if the Wi-Fi went down?

1 Sense · 2 Check · 3 Act locally · 4 Transmit · 5 Store · 6 Present. Talk to your neighbour for one minute.

Who makes it all happen

Group	What they do	Examples
Chip makers	Design the chips and sensors	Espressif, Nordic, STMicroelectronics
Device makers	Build the product people buy	Smart-plug and wearable brands
Connectivity providers	Provide the network	Mobile, LoRaWAN and internet providers
Cloud and platforms	Store data and run rules	AWS, Microsoft, ThingSpeak, Blynk
Developers and integrators	Build apps, connect everything	Software companies, system integrators
End users	Use the system and pay for it	Home owners, farmers, hospitals, cities
Regulators	Set rules on safety, radio, privacy	Telecoms regulators, standards bodies

COMMON MISCONCEPTIONS

Four ideas that trip people up

✗ Myth *An IoT device must connect directly to the internet.*

✓ Fact Many talk only to a nearby gateway, which reaches the internet.

✗ Myth *The cloud does all the thinking.*

✓ Fact More and more decisions happen at the edge: faster, and offline too.

✗ Myth *A Raspberry Pi and an Arduino are the same thing.*

✓ Fact Different families: an OS on a microprocessor vs one program.

✗ Myth *More layers means a more advanced system.*

✓ Fact Three or seven layers can describe exactly the same system.

Map a smart office building

- 1 Draw five boxes: devices, connectivity, edge, cloud, application.
- 2 Put one part of a smart office into each box.
- 3 Which decision should happen at the edge? Why?
- 4 Who sees the result, and what do they see?

KEY TAKEAWAYS

What to remember

- ✓ Five layers: **devices, connectivity, edge, cloud, application**
- ✓ Data flows **up** to people; commands flow back **down**
- ✓ Sensors bring information **in**; actuators push action **out**
- ✓ **Microcontrollers** do one job cheaply; **microprocessors** run an OS
- ✓ A reading follows a **pipeline**, and urgent decisions stay at the edge

Quick quiz

Q1 List the five layers, in order from the device to the person.

Q2 In a smoke detector, is the siren a sensor or an actuator? And the part that detects the smoke?

Q3 A soil sensor must run for a year on a small battery. Which brain: a microcontroller or a microprocessor?

Q4 A smart doorbell stops showing video in its app. Name three places where the fault could be.

UP NEXT • MODULE 3

Sensors, Actuators, and Reading Data

We zoom in on Layer 1: how sensors make their signals, analog vs digital, and your first circuit in Lab 1.



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Book, labs and slides: github.com/datatweets/intro-to-iot-workshop